



Rounding

Accuracy and Precision

- What's the difference?
- According to the clock, the time is 4.03.
- Suppose Person A says, "It is about 4 o'clock."
- Person B says, "It's 9.27."
- Person A's statement is accurate but not precise.
- Person B's statement is precise but not accurate.
- Person C says, "It is 4.03." This is both precise and accurate!



Accuracy and Precision

- We need to be accurate, but how precise do we need to be?
- When using a calculator, we often get more digits than we need.
- For example, the square root of 12 is
3.464101615137755
- We do not usually need this much precision!
- We will consider decimal places and significant figures.



Decimal Places

- The number
 - 52.9628
- Is given to 4 decimal places – there are 4 numerals after the decimal point.
- If we are asked to give the number to 2 decimal places, we drop the third and fourth numerals:
 - 52.96
- But is it always that simple?

Decimal Places

- If the next numeral is 5 or more, we round up.
- If we are rounding
 - 32.1153
- to two decimal places, note that the third numeral after the decimal point is 5, so we round up to
 - 32.12
- 0.38 rounds to 0.40
- 123.4321 rounds to 123.43
- 84.201 rounds to 84.20

Decimal places

- Note that the question will specify how many decimal places.
- The number 3.8392
- is 3.839 to three decimal places,
- 3.84 to two decimal places, and
- 3.8 to one decimal place.

Significant Figures (SFs)

- It is not always useful to round to decimal places.
- For example, rounding
 - 0.0004648
- to three decimal places gives us
 - 0.000
- which is not meaningful!
- We are more likely to round to three significant figures, which gives us
 - 0.000465

Significant Figures

- When rounding to a given number of significant figures, we begin with the first nonzero number.
- Otherwise, the rule is the same as for decimal places: round up if the next numeral is 5 or more.
- Note that sometimes the rightmost figure or figures may be zero.
- The question will typically say how many significant figures. The examples on the next slide are to three significant figures.

Significant Figures

- Rounding to three significant figures:
- 123.2 becomes 123
- 120.2 becomes 120
- 23190 becomes 23200
- 0.05128 becomes 0.0513
- 0.2295 becomes 0.230
- 4990 becomes 5000

General Guide

- When carrying out a series of calculations, do not round off until you get to the final result.
- When multiplying or dividing, the result should be rounded to the same number of SFs as the number with the smallest number of SFs.
- e.g. $12.3456 \times 7.8 = 96.29568$
- 7.8 has 2 SFs, so the answer should be given as 96.
- But “exact” numbers are considered as having infinite SFs.
- e.g. cinema tickets cost £7.73; 2 tickets cost $7.73 \times 2 = £15.46$. As the 2 is exact there is no rounding.